MECHATRONIC SYSTEMS DESIGN LAB

Exercise 3

1. Test the electromagnet provided to you.
2. Write a code to Turn-ON the electromagnet when the proximity sensor senses an obstacle and vice-versa.
3. Calibrate the colour sensor manually. Write a function *ManualCalibrate()* which when called should do the following.
   a. Position the colour sensor over the white patch stuck to the base platform (by rotating base motor).
   b. Run the following algorithm in loop until the user calibrates and enters a stop command (Display the valid stop command. Ex. Press 1 once the calibration is done).
      i. Shine red, green and blue colours with maximum intensity (keep all three potentiometers at its minimum position) individually using tri-colour LED, read the corresponding reflected intensities using LDR and store them in three different variables. (Important - Give a small delay 50-100 ms every time before reading the LDR value)
      ii. Print the read values in the following format in LCD
      R:_____ G:_____ B:_____
      iii. Find the max value (minimum intensity) of the three and adjust the potentiometers to get equal values in all R, G and B (Follow manual calibration procedure explained in class)
      iv. Give a small delay
   c. Once user enters the stop command, exit the above said loop
4. Write a code to auto calibrate the colour sensor. Convert this code to a function *AutoCalibrate()* which when called should do the following.
   a. Set all the potentiometers of RGB LED to its minimal resistance position. **(Note: To be done manually)**
   b. Position the colour sensor over the white patch stuck to the base platform (by rotating base motor).
   c. Read the LDR values by shining R, G and B LEDs individually and store the corresponding reflected intensity values in three different variables.
   d. Find the max value (min intensity) of these three.
   e. Now, vary the intensities of the other two LEDs (analogWrite from 0-255) so that the intensities received from all three are equal.

Note: You may print the status of calibration and the calibrated values in LCD/Serial monitor to understand what is happening at each level

5. Write a program to prompt the user to choose manual or auto calibration and accordingly use the functions to do colour calibration. Once the calibration is done, sweep the base servo within its operating range and display the colour sensed at each step. Display the R, G and B values in LCD.

Expected project demonstration flow

1. Start the program
2. Prompt the user for auto or manual calibration or no-calibration (if calibrated already)
3. Finish calibration and again prompt user for pick up and drop colour location
4. Search for the pick-up colour location using colour sensor (using base motor) and stop once reached.
5. Start the arm motor from its initial position until it senses the object
6. Energize the electromagnet and wait for a pre-defined time (5sec)
7. Retract the arm back to its initial position and search for the destination colour location.
8. Once destination location is reached, bring the arm down, drop the object, retract the arm and go to step 2.

Instructions:

- Save all the codes for evaluation purposes.
- Additional sophistications can be brought in to the project during the demonstration other than what is expected which may have a weightage during evaluation.
- Any similarity in codes between teams will be considered as malpractice.
- All the team members should be able to explain every part of the project during demonstration.